

FOR TEACHERS ONLY

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

PHYSICAL SETTING/EARTH SCIENCE

Wednesday, August 20, 2025 — 8:30 to 11:30 a.m., only

RATING GUIDE

Directions to the Teacher:

Refer to the directions on page 2 before rating student papers.

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: <https://www.nysed.gov/state-assessment/high-school-regents-examinations> and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

Directions to the Teacher

Follow the procedures below for scoring student answer papers for the Regents Examination in Physical Setting/Earth Science. Additional information about scoring is provided in the publication *Information Booklet for Scoring Regents Examinations in the Sciences*.

Allow 1 credit for each correct response.

At least two science teachers must participate in the scoring of the Part B–2 and Part C open-ended questions on a student’s paper. Each of these teachers should be responsible for scoring a selected number of the open-ended questions on each answer paper. No one teacher is to score more than approximately one-half of the open-ended questions on a student’s answer paper. Teachers may not score their own students’ answer papers.

Students’ responses must be scored strictly according to the Rating Guide. For open-ended questions, credit may be allowed for responses other than those given in the rating guide if the response is a scientifically accurate answer to the question and demonstrates adequate knowledge as indicated by the examples in the rating guide. Do not attempt to correct the student’s work by making insertions or changes of any kind. On the student’s separate answer sheet, for each question, record the number of credits earned and the teacher’s assigned rater/scorer letter.

Fractional credit is *not* allowed. Only whole-number credit may be given for a response. If the student gives more than one answer to a question, only the first answer should be rated. Units need not be given when the wording of the questions allows such omissions.

For hand scoring, raters should enter the scores earned in the appropriate boxes printed on the separate answer sheet. Next, the rater should add these scores and enter the total in the space provided. The student’s score for the Earth Science Performance Test should be recorded in the space provided. Then the student’s raw scores on the written test and the performance test should be converted to a scale score by using the conversion chart that will be posted on the Department’s web site at: <https://www.nysed.gov/state-assessment/high-school-regents-examinations> on Wednesday, August 20, 2025. The student’s scale score should be entered in the box labeled “Scale Score” on the student’s answer sheet. The scale score is the student’s final examination score.

Schools are not permitted to rescore any of the open-ended questions on this exam after each question has been rated once, regardless of the final exam score. Schools are required to ensure that the raw scores have been added correctly and that the resulting scale score has been determined accurately.

Because scale scores corresponding to raw scores in the conversion chart may change from one administration to another, it is crucial that, for each administration, the conversion chart provided for that administration be used to determine the student’s final score.

Part B–2

Allow a maximum of 15 credits for this part.

To ensure the accuracy of overlays, select a printer setting such as *full*, *actual size*, or *100%* when printing this document. Do **not** select the *fit to page* setting.

51 [1] Allow 1 credit if *both* responses are correct.

- Metallic luster: magnetite, galena, *or* graphite
- Non-metallic luster: garnet, talc, tourmaline, *or* wollastonite

52 [1] Allow 1 credit for hardness.

53 [1] Allow 1 credit for 50°.

Note: Do *not* allow credit if a compass direction is indicated because this refers to latitude, not an altitude.

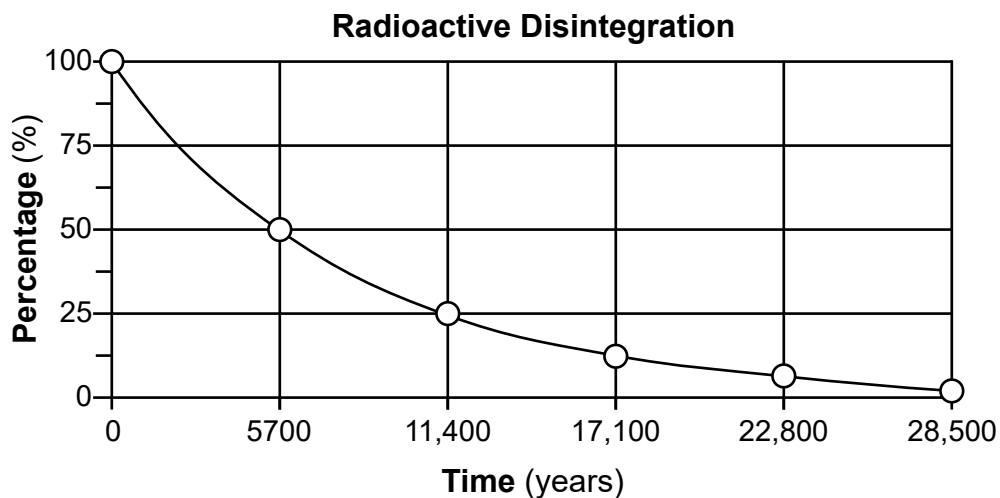
54 [1] Allow 1 credit for 12 hours.

55 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- rotation
- Earth's spinning on its axis.

- 56 [1] Allow 1 credit if *all six* of the student's plots are within or touch the circles shown and all six plots are correctly connected with a line that passes within or touches each circle.

Example of a 1-credit response:



Note: Allow credit if the student-drawn line does *not* pass through the student plots but is still within or touches the circles.

It is recommended that an overlay of the same scale as the student booklet be used to ensure reliability in rating.

- 57 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- Carbon-14
- ^{14}C
- C-14

Note: Do *not* allow credit for " $^{14}\text{C} \rightarrow ^{14}\text{N}$ ", since ^{14}N is neither radioactive nor represented by the data in the data table.

Do *not* allow credit for carbon or C alone because carbon has more than one isotope.

- 58 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- 34,200 years
- 3.42×10^4 years
- 34.2×10^3 years

- 59 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- There would be too little of the radioactive material left after 1 million years.
- The half-life is too short.

60 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- Betelgeuse
- Rigel
- Deneb
- Spica
- Polaris

61 [1] Allow 1 credit for nuclear fusion or fusion.

62 [1] Allow 1 credit if *both* responses are correct.

Name: Milky Way

Shape: spiral or barred spiral

63 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- Contour lines would be farther apart in landscape region *B* and closer together in landscape region *C*.
- The contour lines in landscape region *B* would be spaced farther apart.
- Region *C* would show contour lines closer together.

64 [1] Allow 1 credit for the Hudson River.

65 [1] Allow 1 credit for *G*.

Part C

Allow a maximum of 20 credits for this part.

- 66** [1] Allow 1 credit if *both* relative temperature and relative humidity are correct. Acceptable responses include, but are not limited to:

mT Air Mass:

Moisture:

- high
- humid
- wet
- very moist

Temperature:

- high
- warm/hot

- 67** [1] Allow 1 credit for any value from 5.5 to 7.5 inches.

- 68** [1] Allow 1 credit for 1.6 in/hr.

- 69** [1] Allow 1 credit if *both* responses are correct.

Location A: North American Plate

Location B: Eurasian Plate

- 70** [1] Allow 1 credit for *only* circling “Younger” and an acceptable description of evidence. Acceptable responses include, but are not limited to:

- A is closer to the Mid-Atlantic Ridge and therefore younger than B.
- B is further away from the mid-ocean ridge than A, so it is older than A.
- The age of the ocean floor increases as you move further away from the ridge.
- A is closer to the ridge where the youngest rock is located.

- 71** [1] Allow 1 credit. Acceptable responses include, but are not limited to:
- moving apart/rifting
 - diverging/divergent
 - sea floor spreading
- 72** [1] Allow 1 credit for *E*.
- 73** [1] Allow 1 credit for any value from 29 to 30 days.
- 74** [1] Allow 1 credit for 0.386 million kilometers.
- 75** [1] Allow 1 credit. Acceptable responses include, but are not limited to:
- The Moon’s orbit has an elliptical shape.
 - It is slightly eccentric.
 - almost a circle/circular
 - oval
 - eccentricity of 0.055
- 76** [1] Allow 1 credit if both responses are correct.
- Period: Neogene
 - Epoch: Miocene
- 77** [1] Allow 1 credit. Acceptable responses include, but are not limited to:
- The greater the average global temperature, the greater the height of sea levels worldwide.
 - Higher temperatures lead to higher sea levels.
 - direct relationship/positive correlation
- 78** [1] Allow 1 credit for *two* different greenhouse gases. Acceptable responses include, but are not limited to:
- carbon dioxide *or* CO₂
 - methane *or* CH₄
 - water vapor *or* H₂O gas
 - nitrous oxide *or* N₂O
 - ozone *or* O₃
 - chlorofluorocarbons *or* CFCs

- 79** [1] Allow 1 credit for *only* two check marks — one in the box for “High Pressure” *and* one in the box for “Sinking toward Earth’s surface”.

Note: Allow credit if a symbol other than a check mark is used.

- 80** [1] Allow 1 credit for clay, silt, and sand.

- 81** [1] Allow 1 credit. Acceptable responses include, but are not limited to:

- Water velocity is greater on the outside of a meander.
- The kinetic energy of the river is greater at C.
- The water moves faster near side C of the stream.

- 82** [1] Allow 1 credit for flood plain or plain.

- 83** [1] Allow 1 credit if *both* responses indicate an increase. Acceptable responses include, but are not limited to:

Discharge:

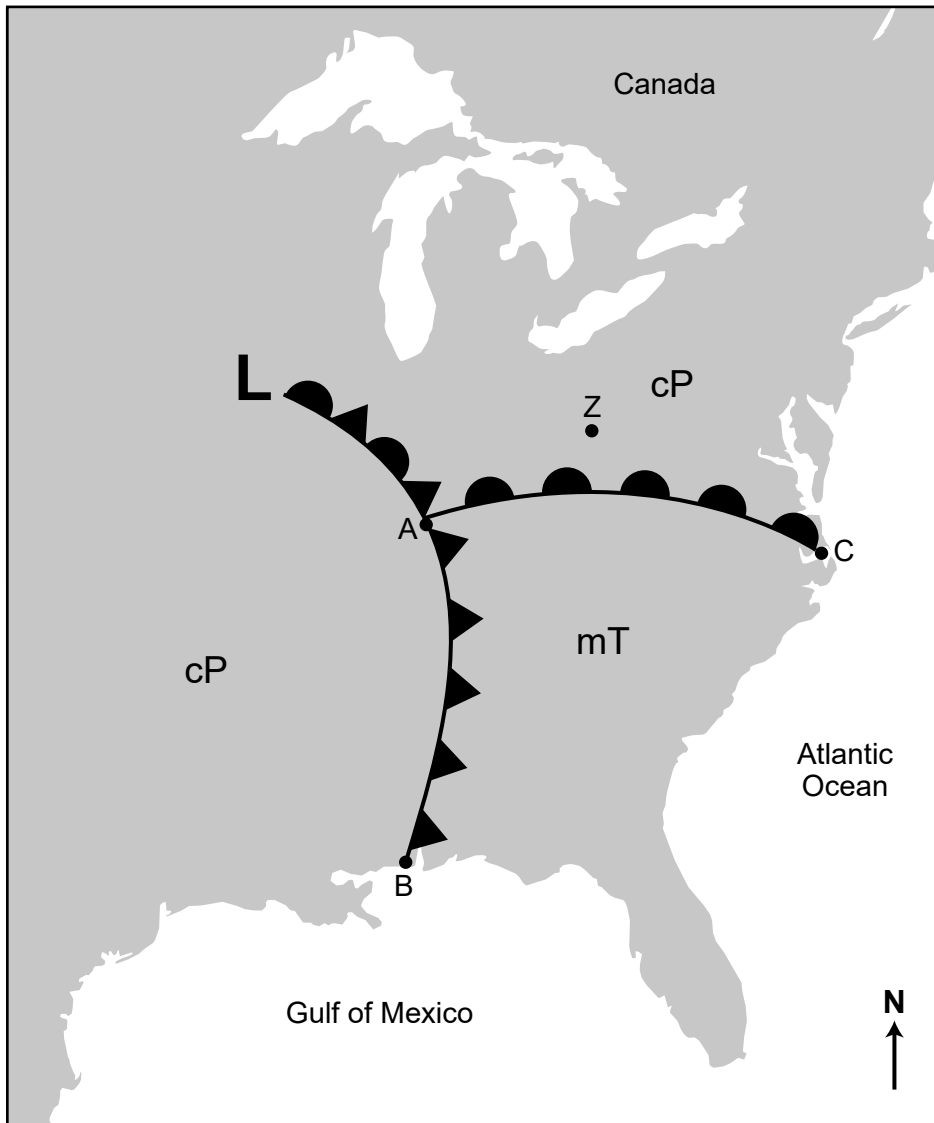
- increases
- is greater
- more volume

Velocity:

- increases
- is greater
- is faster

- 84 [1] Allow 1 credit for correctly drawing a cold front symbol along the length of line *AB* and a warm front symbol along line *AC*. The cold front and warm front symbols must be on the proper sides of the frontal boundaries, as shown below.

Example of a 1-credit response:



Note: Allow credit even if the front symbols are not shaded.

85 [1] Allow credit if *all four* weather variables are correct, as shown in the table below.

Weather Variable	Date
Temperature (°F)	60
Present Weather	rain showers
Wind Direction	— East or E — from the east — to the west
Wind Speed (knots)	any value from 8 to 12

Note: Do *not* allow credit for rain alone for the present weather because the symbol on the model is for rain showers, not rain.

The *Chart for Determining the Final Examination Score for the August 2025 Regents Examination in Physical Setting/Earth Science* will be posted on the Department's web site at: <https://www.nysed.gov/state-assessment/high-school-regents-examinations> on Wednesday, August 20, 2025. Conversion charts provided for previous administrations of the Regents Examination in Physical Setting/Earth Science must NOT be used to determine students' final scores for this administration.

Online Submission of Teacher Evaluations of the Test to the Department

Suggestions and feedback from teachers provide an important contribution to the test development process. The Department provides an online evaluation form for State assessments. It contains spaces for teachers to respond to several specific questions and to make suggestions. Instructions for completing the evaluation form are as follows:

1. Go to <https://www.nysed.gov/state-assessment/teacher-feedback-state-assessments>.
2. Click Regents Examinations.
3. Complete the required demographic fields.
4. Select the test title from the Regents Examination dropdown list.
5. Complete each evaluation question and provide comments in the space provided.
6. Click the SUBMIT button at the bottom of the page to submit the completed form.

Map to Core Curriculum

August 2025 Physical Setting/Earth Science			
Question Numbers			
Key Ideas/Performance Indicators	Part A	Part B	Part C
Standard 1			
Math Key Idea 1	7	42, 56	68
Math Key Idea 2	1, 25, 29, 34	44	67, 73, 77, 80
Math Key Idea 3		49	
Science Inquiry Key Idea 1	3, 4	36, 38, 39, 50, 59	75, 78, 81, 83
Science Inquiry Key Idea 2			
Science Inquiry Key Idea 3	6, 7, 9, 11, 12, 13, 14, 15, 16, 17, 18, 20, 22, 25, 26, 33, 34, 35	37, 38, 45, 47, 51, 52, 57, 59, 60, 64, 65	66, 68, 69, 74, 76, 79, 80, 84, 85
Engineering Design Key Idea 1			
Standard 2			
Key Idea 1	19	43	
Key Idea 2			
Key Idea 3			
Standard 6			
Key Idea 1	8, 10, 21, 28	44	66, 77, 81
Key Idea 2	1, 2, 17, 23, 24, 26, 27, 28, 29, 30, 31, 32, 35	39, 40, 41, 42, 44, 45, 46, 48, 49, 50, 53, 54, 55, 56, 57, 58, 60, 61, 62, 63, 64, 65	67, 68, 69, 70, 71, 72, 76, 79, 80, 82, 84, 85
Key Idea 3			
Key Idea 4			
Key Idea 5	5, 8, 10, 17	55, 56, 65	68, 70, 71, 72, 83
Key Idea 6			
Standard 7			
Key Idea 1			
Key Idea 2			
Standard 4			
Key Idea 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 16, 17, 18, 23, 24	36, 37, 38, 45, 46, 48, 49, 50, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62	72, 73, 74, 75, 76, 77
Key Idea 2	10, 11, 12, 13, 14, 15, 19, 20, 21, 25, 26, 27, 28, 29, 30, 31, 32	39, 40, 41, 42, 43, 44, 63, 64, 65	66, 67, 68, 69, 70, 71, 78, 79, 80, 81, 82, 83, 84, 85
Key Idea 3	22, 33, 34, 35	47, 51, 52	
Reference Tables			
ESRT 2011 Edition (Revised)	6, 7, 9, 11, 12, 13, 14, 15, 16, 17, 18, 20, 22, 25, 26, 33, 34, 35	37, 38, 45, 47, 51, 52, 57, 59, 60, 64, 65	66, 68, 69, 74, 76, 79, 80, 84, 85